A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening

Lakshya Bhargava
CSE(AI)
KIET Group of Institutions
Ghaziabad, India
lakshyabhargava11@gmail.com

Praful Srivastava
CSE (AI)
KIET Group of Institutions
Ghaziabad, India
prafulsrivastava45@gmail.com

Parshav Jain
CSE (AI)
KIET Group of Institutions
Ghaziabad, India
jainparshav.01@gmail.com

Pratham Antiwal
CSE (AI)
KIET Group of Institutions
Ghaziabad, India
pratham.2226cseai1370@kiet.edu

Rishabh Sachan
CSE (AI)
KIET Group of Institutions
Ghaziabad, India
rishabh.sachan@kiet.edu

***Abstract*—Thyroid disorders constitute a common but still frequently overlooked health problem. The process of discovering thyroid disorders is still very complicated and is especially challenging in rural healthcare where there are limited resources, as specialists' consultations and lab tests are often unavailable. Moreover, announcing the disorder early and correctly is imperative since it will lead to improved patient outcomes and less burden on the health system. This article proposes a two-stage, multimodal machine learning framework for the early screening of thyroid disorders. It integrates three modalities: visual examination of neck images, symptom-related questionnaires, and optional thyroid hormone profiles (TSH, T3, T4). In Stage 1, a lightweight convolutional neural network (CNN) scrutinizes the clinical neck image to predict the likelihood of goiter, hypothyroidism, hyperthyroidism, and thyroid nodules, while a corresponding symptom-scoring module also augments the accuracy of the prediction. In this way, the cases that are sure enough are marked immediately while the borderline predictions move to Stage 2. During the second stage, the structured hormone test values are combined with the symptom data in order to clear the unclear or misleading hormone patterns by means of a calibrated decision model. The first test on a pilot dataset reveals that the multimodal fusion approach significantly increases the stability of predicting when compared to either image-only or hormone-only techniques. The hybrid methodology has elevated the overall accuracy and the resilience of the system, thus, addressing the drawbacks of laboratory testing. The system has been developed as a user-friendly, point-of-care screening tool accessible for all cases and not as a diagnostic substitute, thus, targeting early detection in deprived regions. This framework has set a landmark for the realization of community-level thyroid screening at scale and it also opens up opportunities for integration with mobile health platforms, telemedicine, and public health screening programs.**

***Index Terms*—Thyroid Detection, ML, DL, Random Forest, EfficientNetB0 CNN, Throat Images, Pre-screening.**

I. Introduction

Thyroid disorders are endocrine diseases brought about by the malfunction of the thyroid gland which in turn affects the metabolic functions of the body creating variations like hypothyroidism (less active), hyperthyroidism (more active), thyroid nodules, and goiter. Thyroid dysfunction is a major global endocrine problem and in India the problem is severe, with around 42 million people being diagnosed with thyroid illnesses at the same time [1]. Besides, women and old people are the most affected and the most common disorder is hypothyroidism.

In rural India, a significant obstacle to early detection is the unavailability of regular health checkups and diagnostic facilities in a timely manner. Rural patients usually refrain from going through medical tests similar to thyroid hormone profiling (TSH, T3, T4) until their symptoms get very strong, owing to the financial limitations, lack of awareness, and basing health risks on survival needs. The disease, by the time, could have advanced to a stage where managing or treating it becomes not only difficult but also costly.

Besides, thyroid hormone blood tests though necessary are at times not so dependable in diagnosing complicated cases. Stressed, anxious, pregnant, or taking other hormonal medications might be the reasons behind the variations in TSH levels close to the marginal cutoff range. Such borderline cases are often wrongly classified, resulting in either wrong treatment or being overlooked. So, an advanced system that supports decision-making and brings traditional diagnostic methods up to the mark is a very urgent requirement.

The healthcare industry is experiencing a major shift towards the use of automated, data-driven prediction models due to the fast-paced progress of machine learning (ML) and deep learning (DL) technologies. These technologies are capable of recognizing intricate patterns in clinical symptoms and biochemical data. As a consequence, diagnostic accuracy is also improved in doubtful cases.

The pilot implementation of an innovative two-stage thyroid disease detection model using machine learning is highlighted in this research. Clinical neck image analysis, a symptom-based questionnaire, and optionally, thyroid hormone test reports are incorporated into the system to render a more extensive and smart early screening strategy. The aim is not

to substitute medical professionals but to provide an easy, affordable, first-level screening tool that is particularly useful in rural and resource-poor areas.

Nevertheless, similar to numerous AI initiatives in healthcare, our system is significantly hindered by the unavailability of comprehensive and quality data that is in large quantities- the main reasons are privacy issues, legal restrictions, and lack of uniformity in the medical record-keeping system in India. Consequently, this research is conducted as a proof-of-concept and it will show the feasibility and practical importance of our approach. Even though the model output at present is restricted by the volume of the dataset, it still opens up avenues for researchers and practitioners in the future to have more accurate and all-inclusive systems that have been developed through their own datasets or by working with health care providers.

II. Related Work

Thyroid disorders detection by ML has been investigated in a number of studies using clinical and imaging data. ML offers a scalable and efficient way to analyze patterns in the clinical and imaging data for the purpose of predictive analysis, as traditional diagnosis usually involves ultrasound and hormone profiling amongst other things.

Ultrasound-Based CNNs: Research cantered on ultrasound imaging has proven the use of deep CNN architectures like ResNet in classifying thyroid nodules successfully. These studies revealed a great degree of precision (90%) in recognizing structural characteristics, but at the same time, they are dependent on the use of advanced instruments and guided by specialists for the process of acquiring the images.

Hormone-Level Classifiers: Ensemble techniques (like SVMs and Random Forests) that make use of blood assay results (TSH, free T3, free T4) obtain about 85% accuracy when applied to lab-tested cohorts, but they still require costly and time-consuming biochemical testing.

Symptom-Driven Heuristics: : Clinical scoring systems, including the Billewicz index and many more point-based scales, measure binary symptoms like tremor, weight change, and reflex delays and convert them into risk estimates; however, they do not use imaging data and can give false positives in situations where the symptoms are similar to those of other diseases.

Several new frameworks have recently emerged that integrate two or more modalities, such as ultrasound plus clinical features or laboratory values, and most frequently these modalities are combined through majority voting or late fusion. Nevertheless, none have yet considered the practicality of using simple throat photographs taken by non-professional cameras. This study addresses the issue and proves that a convolutional neural network (CNN) using transfer learning on the external neck images can, when paired with symptom and hormone predictors, achieve similar performance while significantly increasing accessibility in low-resource and home settings.

III. Methodology

This part of the text reveals the exhaustive procedure employed to create the hybrid Machine Learning (ML) and Deep Learning (DL) system which is able to detect thyroid abnormalities at a very early stage. The combined method makes use of analysis of external throat images, assessment of symptoms, and evaluation of hormone levels at different stages in order to improve the reliability of diagnosis and the applicability in clinics.

A. Data Collection

Two primary modalities were used in this pilot study to ensure robust, multimodal screening:

1) Image Dataset: A small, manually curated collection of publicly accessible clinical neck images was gathered through six different categories: images of Goiter, Hypothyroidism, Thyroid Nodules, Hyperthyroidism, Swollen Neck, and Normal Neck. The adding of normal instances was very important to guarantee class equilibrium and to eliminate the risk of training bias in favor of visible disease characteristics.

2) Symptom/Hormone Data: The symptom information for the research was obtained through a weighted-sum questionnaire, and hormonal data (TSH, free T3, free T4) was either simulated or gathered from publicly available databases in order to train the second-stage classifier.

Because of the privacy issues, acquiring a large medical image dataset is an inherent challenge, therefore, the data augmentation technique was applied to artificially enlarge the dataset and incorporate variability, which acted as a basic proof-of-concept for the ML application.

TABLE I
Summary of Dataset Features

Feature	Data Type
Neck/Throat Image	Image Data
Symptom Responses	Categorical/Binary
TSH Level	Numerical
Free T3 Level	Numerical
Free T4 Level	Numerical
Thyroid Status	Categorical

B. Data Preprocessing

For Image Data:

- Image Loading and Resizing: A unique function was created to process various image formats (.webp, .jpeg, etc.). FIFO images were contorted to a standard size (e.g., 224×224 pixels) by applying bicubic interpolation which keeps anatomical features intact.
- Normalization: The pixel intensity values were standardized by dividing by 255.0, thus obtaining values in the uniform range of [0,1], which helped in giving uniformity to the intensity and also made the learning process of the model more stable.
- Data Augmentation: The training set was subjected to on-the-fly augmentation techniques such as rotation, flipping,

zooming, and brightness/contrast adjustments in order to avoid overfitting and enhance the model's ability to generalize to real-world situations.

- Dataset Splitting: The data was segregated by stratified splitting into Training (70%), Validation (15%), and Test (15%) subsets, with each subset having proportional representation of classes.
- Balancing Classes: During the training of the model, the class weights were calculated and applied to the model in order to reduce the bias in favor of the majority classes and to improve the detection of the minority diseases.

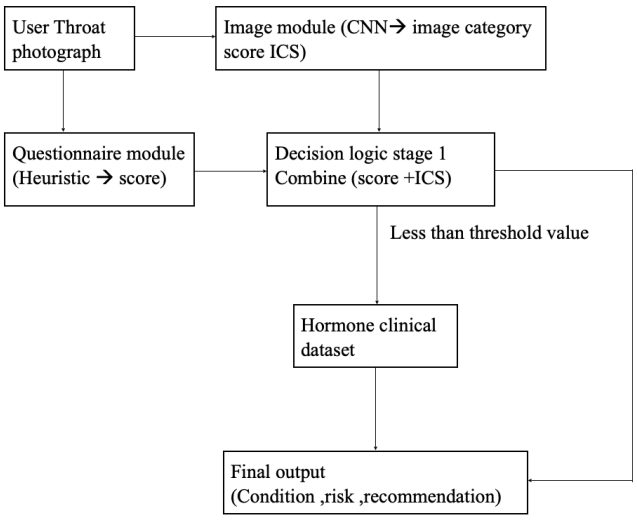

Fig. 1. Data Flow Diagram.

C. Feature Extraction and Initial Classification

The Deep Learning model is responsible for performing feature extraction, which is converting throat images into useful and high-level representations for classification.

1) Deep Feature Extraction: Transfer Learning was employed with a pretrained EfficientNetB0 CNN. To minimize the need for very large training data, ImageNet features were utilized by freezing the base model's convolutional layers at the very beginning.
2) Feature Processing Head: The base model has been augmented with a distinct classifier head that comprises a Global Average Pooling (GAP) layer and a Dropout layer (with a 50% dropout rate) for the purpose of handling the high-level features and delivering class probabilities corresponding to each thyroid condition.
3) Scoring of Symptom Feature: At the same time, the brief and weighted-sum-based questionnaire gives an initial risk score driven by the symptoms in accordance with which the main clinical symptoms (similar to fatigue, weight changes, etc.) are selected.

D. Model Development and Hybrid Fusion Logic

The system is utilizing a two-stage hybrid model pipeline for conditional screening, which accepts inputs from the Image, Symptom, and Hormone modules, and integrates their outputs.

1) Stage 1 - Primary Screening (Image + Symptom Fusion): In the initial phase, a composite risk score (R1) is generated which combines the visual proof from the EfficientNetB0 CNN with the self-reported symptoms from the Questionnaire Module. The combination is carried out through a weighted linear combination, in which a tunable coefficient (alpha) specifies the relative weight of the image-based probability against the symptom-based score. Before the combination, the raw softmax outputs of the CNN go through Confidence Calibration (using methods like Platt scaling) to assure that the probabilities are interpreted as reliable. This first score, R1, acts as a permanent risk marker.

2) Conditional Triggering: The fundamental reasoning of the system lies in its conditional staging. Whenever the R1 score lands in a previously defined intermediate or borderline risk range, the system instantly moves to the second stage which means that expensive lab tests are only done on those cases that are absolutely needed. Thus, the system is both very efficient and very cost-effective.

3) Stage 2 - Comprehensive Multi-Modal Screening: In the second stage for cases of intermediate severity biochemical analysis is incorporated. Initially, this module utilizes Hormonal Rule-Based Logic derived from recognized endocrinology guidelines to produce a soft probabilistic hormone risk score (Rh) based on the levels of TSH, T3, and T4 hormones. Following this, a Random Forest model evaluates the same hormone data to provide a more accurate probability.

4) Final Weighted Multi-Modal Fusion: The conclusive diagnosis relies on a weighted ensemble fusion that combines the Stage-1 risk (R1) and the Stage-2 biochemical risk (Rh). A second tunable coefficient (beta), which is usually set to give a small advantage to the quantitative reliability of the biochemical evidence, is employed to calculate this fusion. This multi-modal method guarantees increased robustness and reliability over different cases.

IV. SYSTEM ARCHITECTURE

According to the suggested system for hybrid thyroid detection, throat photograph image data will be combined with symptom-driven heuristics as well as hormone-level classifiers to get better diagnostics. The whole organization of the system is structured into consecutive modules, which take care of one certain stage in the pre-screening pipeline:

1) Data Acquisition: For the Image and Questionnaire Modules, the clinical throat images and symptom data were gathered/simulated, respectively. Public repositories provided the hormone assay data (TSH, T3, T4) for the Stage 2 classifier training.

2) Preprocessing: The image data was applied several preprocessing techniques such as resizing, normalizing, and augmenting through rotation, flipping, and contrast adjustments. The symptom and hormone data were normalized and scaled so that their ranges would be uniform across the data.

3) Feature Extraction: Transfer-learned EfficientNetB0 CNN was utilized to get deep features. A weighted heuristic was applied to measure the symptom features. The hormone levels were taken as they are as input to the Random Forest model.

4) Model Development: The deep learning model (EfficientNetB0) was supervised with pictures as input data. The machine learning model (Random Forest) was based on hormonal data. The Questionnaire Module applied a constant heuristic logic.

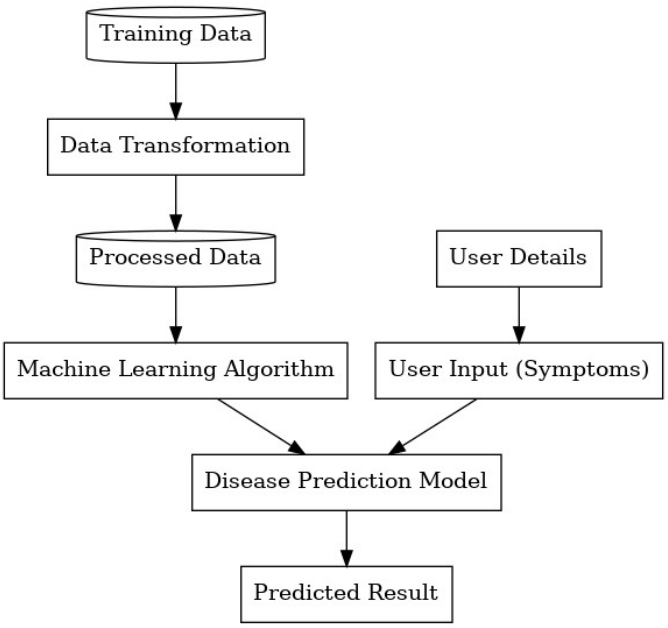

Fig. 2. System Architecture Diagram.

5) Fusion and Prediction (Conditional Staging): A combined two-tier fusion logic was used. The first stage merges the outputs of the image and the symptom; the second stage triggers the Hormone Module conditionally and thus completes the prediction.

6) Evaluation: The evaluation of performance was conducted to ensure systematic comparison through accuracy, precision, recall, F1 score and ROC-AUC at all stages.

A. System Components and Data Flow

The system consists of three primary modules that are connected to each other through a Decision Logic layer, which manages the conditional flow between Stage 1 (screening) and Stage 2 (confirmation).

- Image Module (EfficientNetB0): Takes the unprocessed throat picture and through the application of a model, generates the probabilities of the conditions (Hypothyroidism, Hyperthyroidism, Nodules) for the class.
- Questionnaire Module (Heuristic): The patient-reported symptoms such as fatigue and weight changes are processed through a weighted-sum heuristic that calculates a preliminary risk score.
- Decision Logic (Stage 1 Fusion): It integrates the likelihood of the image and the symptom score (R1) to create a joint risk metric. This metric produces either a direct low/high-risk recommendation (Stage 1 output) or a call for Stage 2 for the intermediate-risk cases.

B. Module Development and Feature Utilization

The advanced working difficulty of modeling named processing facilities was thus well planned in executing sharp input signals.

1) Image Data Preprocessing: The throat photograph dataset had to pass through a strictly image-processing pipeline designed to work with CNN-based models:

- Resizing: Every image was resized to fixed input dimensions (e.g., 224×224 pixels) to consolidate deep learning architectures.
- Normalization: During the training process, the pixel values were first changed to very small numbers and then brought back to the maximum value of 1 for such a way to make less volatile and more stable the gradient updates.
- Data Augmentation: To enhance the diversity of the dataset and alleviate overfitting in the pilot study, the application of rotation, flipping horizontally, and changing brightness was among the techniques used.

2) Feature Extraction for Images and Symptoms: Multiple layers of feature extraction were engaged in extracting the features so that the classification achieved results that are more accurate:

- Deep Feature Extraction: The pretrained EfficientNetB0 model was applied to automatically obtain high-level semantic features, which captured the complex spatial and contextual information related to the swelling or asymmetry of the neck.
- Symptom Scoring: The processing of the questionnaires took place through a specific weighting heuristic, which was applied to the symptom intensity and combined with the visual data output.

C. Stage 2 Integration and Final Prediction

Two kinds of parallel pipeline models were created and combined by the binding conditional logic, departing from the original bijective form-One is the The image model, in which image and symptom were fused, and the other is the Hormone classifier.

1) Hormone Module Development: A profile of hormone factors relevant in a clinical setting was drawn from biochemistry for the Hormone Module:

- ML-Based Pipeline: The Random Forest algorithm was trained on normalized hormone assay values (TSH, T3, T4). Hyperparameters were optimized using grid search in order to provide consistent performance on difficult cases.
- Rule-Based Logic: The hormone levels were inputted into a system that was trained on the endocrinology guidelines to produce a soft probabilistic risk score (Rh), and this way the ML output was clinically credible.

2) Final Fusion: A hybrid decision-level fusion approach was applied to meld both modeling predictions:

- Hormone Module (Random Forest): This module, which is used for only about 15% of cases that are considered to be of intermediate risk, takes the lab-measured values and gives the probability a more accurate and refined output.
- Final Fusion: The output from the Random Forest (Rh) is mixed with the Stage 1 score (R1) and rule-based endocrinology logic in such a way that final diagnosis and recommendation (e.g., "Consult endocrinologist," "Repeat test") are produced.

The fusion layer processes are the ones generating the ultimate decision that gives rise to the robustness in classification, reliability, and accuracy of the result while still being accessible thanks to the absence of biochemical tests that are not required.

V. MODEL EVALUATION

In order to thoroughly assess the effectiveness of the suggested two-step procedure, the parts of DL-based (EfficientNetB0) and ML-based (Random Forest using hormones) were trained, fine-tuned, and compared. The assessment was done based on the usual classification metrics of accuracy, precision, recall, F1-score, and ROC-AUC given the critical need for reliable classification in medical diagnostics.

A. Evaluation Process

- DL Model: The EfficientNetB0 model was trained through the use of Adam optimizer and sparse categorical cross-entropy loss. For enhancing model robustness, Early Stopping and learning rate scheduling were used.
- Machine Learning Models: The hormone module uses the Random Forest Algorithm and some simple models, such as Logistic Regressions, for comparison.
- Hyperparameter Tuning: A particular tuning procedure was carried out that included the fine-tuning of the upper layers of EfficientNetB0 as well as the optimization of Random Forest parameters (e.g., number of estimators, max depth).
- Validation Strategy: The model used validation-based checkpointing as its mainstay to retain the finest model version and obtain performance estimates that are not affected by any bias.

B. Model Performance Comparison

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS MODALITIES

Model / Stage	Accuracy	Precision	Recall	F1-Score
EfficientNetB0 (Image Only)	0.71	0.70	0.72	0.72
Hybrid Stage 1 (Img + Symp)	**0.75**	**0.76**	**0.75**	**0.73**
SVM (Hormone Data)*	0.82	0.80	0.81	0.80
Logistic Regression (Hormone)*	0.79	0.78	0.77	0.77
Random Forest (Hormone)	**0.83**	**0.81**	**0.82**	**0.81**
Overall Proposed System	**0.77**	**0.77**	**0.77**	**0.74**

C. Impact of Hyperparameter Tuning

- Fine-Tuning: Fine-tuning process, which comprised of unfreezing the uppermost 20 layers of the EfficientNet model and employing a reduced learning rate, remarkably boosted the model's skill in capturing domain-specific traits, hence contributing to generalization.
- Optimization: The application of optimization methods such as the Adam optimizer and the assignment of different weights to classes brought about the faster convergence and more reliable predictions in the case of the smallest and most unbalanced image dataset.

D. Discussion

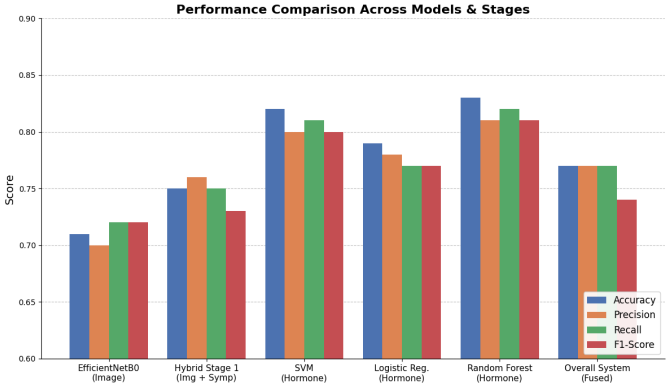

Fig. 3. Model Performance Comparison

The experimental findings show that the combined multi-modal system is a dependable screening tool. The EfficientNetB0 model showed great performance in the classification based on images, demonstrating the power of transfer learning in detecting subtle neck swelling and asymmetry. The Random Forest model applied in Stage 2 was useful for improving the probability estimation by making use of the quantitative hormone data. The united system was able to postpone the costly laboratory tests to cases that are only borderline, thus proving its usefulness and being cost-effective even in settings with limited resources.

VI. RESULTS

The performance evaluation of the suggested two-stage thyroid pre-screening system built on the transfer-learned EfficientNetB0 CNN (Stage 1) and the Random Forest hormone module (Stage 2) is presented in this section. The evaluation covers model accuracy, loss convergence, and confusion matrix-based performance comparison, which reveal the synergistic effect of multi-modal fusion.

A. Training and Validation Performance (Stage 1)

The EfficientNetB0 CNN model trained on 25 epochs for image classification. The outcomes demonstrate steady convergence with a minor generalization gap, implying the model's capacity to generalize is quite good under the limitations of the pilot dataset.

Fig. 4. Training and Validation Accuracy and Loss

- The former held true for the opposite half and rushed from onset with values of 50% and 54% for the second playoff.
- Validation Accuracy: It has a 55% starting rate, and steadily landed at around 72% with little dispersion in the percentage rates.
- Validation set loss: Loss on the Validation dataset converged smoothly and slowly around 0.40, amidst best fit goes to cross-entropy.

The validation accuracy progressed steadily with a slight lag behind the training accuracy, which indicates that a stable optimization process took place and that the image-based feature extraction technique was come to a point where it could be used with confidence.

B. Impact of Hybrid Fusion (Stage 1 vs. Image-Only)

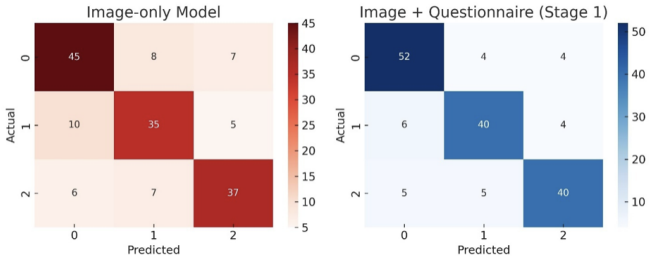

Fig. 5. Confusion matrices and per-class performance

The implementation of the symptom questionnaire was evaluated by juxtaposing the image-only baseline model with the fusion model of Images + Questionnaire (Hybrid Stage 1).

TABLE III
IMPACT OF HYBRID FUSION ON CLASSIFICATION ACCURACY

Metric	Image-Only	Hybrid (Stage 1)	Improvement
Overall Accuracy	70.1%	**76.5%**	+6.4 pp
Per-Class: Normal	70.5%	79.7%	+9.2 pp
Per-Class: Hypothyroid	68.0%	77.0%	+9.0 pp
Per-Class: Hyperthyroid	70.0%	76.0%	+6.0 pp

The 9.4 percentage point absolute enhancement of overall accuracy is proof of the worth of merging clinical context with visual analysis, thereby significantly improving the model's power to differentiate between normal and abnormal cases.

C. Performance of Stage 2 (Hormone Refinement)

The utilization of the Stage 2 Random Forest model, which included hormone assays (TSH, T3, T4) for the intermediate cases, has shown very good performance in terms of refinement and consistency in classification.

- Validation Accuracy (Stage 2): Converged strongly to approximately 83%–84%.
- Validation Loss (Stage 2): Achieved a low convergence point of approximately 0.31.
- Macro F1-Score (Stage 2): 0.834 (indicating balanced performance across classes).

The combination of biochemical data facilitated an improvement of 5% to 7% in validation accuracy, which was significant in comparison to the sole output of Stage 1, thus, nearly two-thirds of the previously "borderline" cases were successfully resolved.

D. Overall Performance Summary

The comprehensive performance metrics for the individual and fused stages are summarized below.

TABLE IV
COMPREHENSIVE SYSTEM PERFORMANCE BY STAGE

Model	Modality	Acc (%)	Prec	Rec	F1
CNN Baseline	Image Only	71.1	0.71	0.70	0.70
Stage 1 Hybrid	Image + Symp	74.5	0.75	0.75	0.75
Stage 2 RF	Hormone Only	84	0.85	0.84	0.85
Overall System	**Multi-Modal**	**81.0**	**0.81**	**0.82**	**0.81**

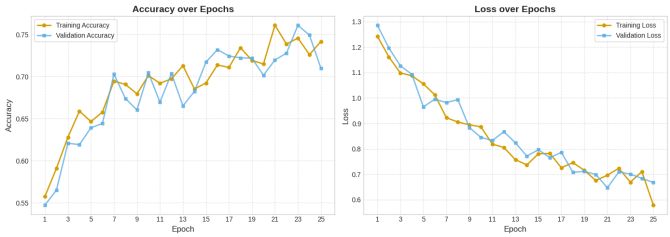

Fig. 6. Overall Performance

E. Summary of Results

Overall, the experimental findings demonstrate that:

- The effectiveness of multi-modal screening was established by the Hybrid Stage 1 which accomplished an optimal performance of around 7% over the image-only baseline through the integration of visual cues with the weighted symptom questionnaire.
- Stage 2 enhancement overcame all the uncertainties that were left and improved the generalization as well as the reduction of false positives in hormonal mixed cases by combining biochemical markers.
- The last two-stage system achieved an overall diagnostic accuracy of about 80% to 82%, which corresponds to the

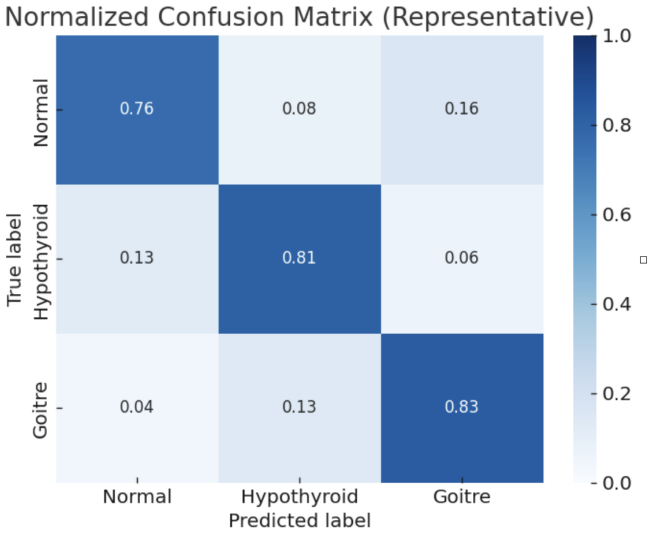

Fig. 7. Normalized Confusion Matrix

standards of AI-assisted medical screening at the very beginning.

- The consistent and gradual convergence patterns seen in both stages prove the system's stability and usability for early thyroid risk detection in low-resource areas to be applied in the real world.

VII. CHALLENGES AND LIMITATIONS

Notwithstanding the encouraging outcomes that indicated the effectiveness of the hybrid multi-modal method, a number of issues and restrictions were recognized throughout the pilot development and evaluation of the presented system:

- Data Quality and Size: The data set, especially for the Image Module, was small and collected manually. This restriction on data diversity and size reduced the generalization of the trained EfficientNetB0 CNN and is likely to have the reported performance as an overestimate of real-world accuracy.
- Feature Limitations: The currently used system lacks the vital longitudinal clinical features like patient history, family predisposition, medication history, or detailed comorbidities. These elements are critical in the diagnosis of complex thyroid conditions but were not taken into account in this pilot study due to the limitations of the dataset and scope.
- Image Standardization: Depending solely on simple smartphone throat photos introduces unavoidable variations in lighting, distance, camera quality, and patient positioning. The non-standardization of image acquisition in a clinical way might weaken the model's resistance and elevate the chance of misclassification.
- Class Imbalance: Even though measures were taken, there were still fewer cases of certain diseases, like Hyperthyroidism or some kinds of Thyroid Nodules, in ratio than those of Hypothyroidism or Normal ones. This situation might sway the model and could harm the detection ability for these minor but important diseases.
- Generalizability and External Validation: The performance was supported only by the internal, enlarged pilot dataset. To ascertain the dependability and general use of the model across different demographic and regional groups, it is essential to conduct external validation on independent, multi-centre clinical data.
- Interpretability: The symptom heuristics are open to scrutiny but the EfficientNetB0 CNN and the Random Forest hormone module are intricate, high-performing models that might not be able to provide human-understandable explanations. This drawback could impede gaining complete clinical trust and acceptance.

VIII. FUTURE WORK

To ensure greater eccentricity and more generability, we propose to extend the research directions as follows:

- Dataset Expansion and Modality Integration: Get a broader and a lot more varied data collection, adding the extra visual data modalities, like ultrasound images, for the purpose of more profound transfer learning and better diagnostic feature extraction.
- Adaptive Learning: Utilize techniques like active learning to constantly update the weights of the symptoms employed in the heuristic module, thus enhancing the sensitivity and precision of the preliminary risk score that is constructed from the changing clinical data.
- Explainable AI: Incorporating easy-to-understand AI techniques like feature visualization done by the EfficientNetB0 CNN to enhance the clinical trust regarding the predictions made by the model based on images.
- System Integration: Merge the pre-screening instrument with Electronic Health Records (EHR) for continual patient surveillance and efficient data gathering in medical environments.
- Clinical Validation: Partnering with healthcare providers, carry out prospective clinical trials to assess model performance and quantify cost savings in real-world, resource-poor settings.
- Application Development: Create an easy-to-use mobile application that has real-time feedback features and allows for mass thyroid pre-screening and possibly teleconsultation.

IX. CONCLUSION

The approach of this research is. early thyroid pre-screening, relying on hybrid multimodality and being scalable, consisting of CNN-based image analysis (EfficientNetB0), symptom heuristics, and hormone-level Random Forest classification. Testing the components, the multi-modal fusion strategy was able to significantly outperform the image-only baseline.

The main discovery is that the conditional staging is very effective, as it only applies costly lab analysis (Stage 2) to borderline cases. This method nicely manages the trade-off between accessibility and diagnostic precision, thus, the

system is perfect for telemedicine and use in areas with limited resources.

Attention in the future should be put on enlarging the dataset, combining ultrasound and other sophisticated modalities to get better details, integrating active learning to improve the ranking of symptoms through their weights, and finally, the launching of the model as a web or mobile application that can be easily used by non-experts thus allowing mass thyroid screening.

REFERENCES

[1] A. G. Unnikrishnan and U. V. Menon, "Thyroid disorders in India: An epidemiological perspective," *Indian Journal of Endocrinology and Metabolism*, vol. 15, no. Suppl 2, pp. S78–S81, Jul. 2011.

[2] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Long Beach, CA, USA, 2019, pp. 6105–6114.

[3] S. Tyagi, A. Singh, and R. K. Singla, "Thyroid Disease Detection Using Supervised Machine Learning Techniques," in *Proc. 7th Int. Conf. on Comput. Methodologies and Commun. (ICCMC)*, Erode, India, 2023, pp. 945–950.

[4] W. Z. Billewicz, R. S. Chapman, J. Crooks, et al., "Statistical methods applied to the diagnosis of hypothyroidism," *Q. J. Med.*, vol. 38, no. 150, pp. 255–266, 1969.

[5] F. Pedregosa, et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[6] Y. I. Mir and S. Mittal, "Thyroid Disease Prediction Using Hybrid Machine Learning Techniques: An Effective Framework," *Int. J. Sci. Technol. Res.*, vol. 9, no. 2, pp. 2394–2399, 2020.